A. GENERAL NOTES

A1. STRUCTURAL DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL OTHER RELEVANT DRAWINGS AND THE SPECIFICATION.

A2. FOR MAIN SETTING OUT AND LEVELS REFER TO ARCHITECTURAL DRAWINGS

A3. ALL ELEMENTS OF THE STRUCTURE SHALL BE DESIGNED AND VERIFIED BY THE CONTRACTOR SHALL BE SUBMITTED TO THE CONSULTANT FOR APPROVAL.

A4. THE CONTRACTOR SHALL PROVIDE TEMPORARY BRACING AND SHORING AGAINST LATERAL FORCES AND ALL CONSTRUCTION LOADS THROUGHOUT THE CONSTRUCTION PROCESS.

A5. THE WORKS SHALL BE CARRIED OUT WITH DIMENSIONAL LIMITS THAT ARE SUITED TO THE STRUCTURAL STABILITY & FINAL APPEARANCE.

A6. LEVELS INDICATED ARE IN METRES. REFER TO ARCHITECT'S DRAWINGS FOR RELATIONSHIP TO DUBAI MUNICIPALITY DATUM (DMD).

A7. LOADINGS FOR THE BUILDING ARE:

(i) WIND LOADING IN ACCORDANCE WITH ASCE 7-16
BASIC WIND SPEED, V (3 SECOND GUST) = 45 m /Sec.
WIND LOAD FACTORS :
EXPOSURE TYPE : C (OPEN TERRAIN)
IMPORTANCE FACTOR, I : 1.00
TOPOGRAPHICAL FACTOR, K_{zt} : 1.00
GUST EFFECT FACTOR, G : 0.98
DIRECTIONALITY FACTOR, K_d : 0.85
PRESSURE COEFF, C_p : 0.50 (LEEWARD)
: 0.80 (WINDWARD)

(ii) SEISMIC LOADING IN ACCORDANCE WITH THE ASCE 7-16
0.2 SEC. SP. ACEL. S_s 0.51
0.2 SEC. SP. ACEL. S_1 0.18
SITE CLASS-C

OCCUPANCY CATEGORY IV STANDARD OCCUPANCY

NOT WITHSTANDING THE ABOVE THE CONTRACTOR SHALL CARRY OUT ANY ADDITIONAL TESTING HE DEEMS NECESSARY TO ENSURE SATISFACTORY PERFORMANCE OF ALL ELEMENTS INCLUDES THE CLADDING SYSTEM.

(iii) DEAD AND IMPOSED LOADING IN ACCORDANCE WITH ASCE 7–16 EXCEPT AS NOTED BELOW: DEAD LOADS ARE TO BE CALCULATED FROM THE UNIT WEIGHTS GIVEN IN RELAVANT CODES OR FROM THE ACTUAL KNOWN WEIGHTS OF THE MATERIALS USED. THE UNIT WEIGHT OF CONCRETE IS NOT TO BE TAKEN AS LESS THAN 25kN/m3.

IN ACCORDANCE WITH THE REQUIREMENTS OF AUTHORITY LIVE LOADING FOR CAR PARKING AREAS IS TO BE 3.5 kN/m^2 (MIN.)

A8. PRIMARY ELEMENTS TO BE DESIGNED BY THE CONTRACTOR ARE AS FOLLOWS: (WHEREVER APPLICABLE)
PRECAST CONCRETE FLOOR UNITS, CLADDING IN ALL TYPES, CURTAIN WALLS,
POST TENSIONED CONCRETE FLOORS, SLAB AND BEAMS
ARCHITECTURAL FACADE AND SUPPORT STEEL WORK, ARCHITECTURAL ROOF FACADE AND SUPPORT STEEL WORK STEEL STRUCTURE SHALL BE SUBMITTED FOR ENGINEERS / AUTHORITY APPROVAL AS REQD.

A9. STRUCTURE DESIGN LIFE = 50 YEARS.

A9A. FIRE RESISTANCE REQUIREMENT FOR STRUCTURE = 2 hrs. (MIN) AND SHELL BE AS PER AUTHORITY RELEVANT CODES & GUIDE LINES.

A10. ALL ASTERISKED DIMENSIONS (INDICATED THUS: 98*) ARE TO BE CONFIRMED ON SITE AND REPORTED TO THE CONSULTANT FOR CONSIDERATION.

A11. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS NOTED OTHERWISE.

A12. BUILDING MOVEMENTS- CONTRACTOR'S METHOD STATEMENT FOR CONSTRUCTION SHOULD CONSIDER ALL BUILDING MOVEMENTS ARISING FROM DIFFERENTIAL LOADING,CREEP,SHRINKAGE, THERMAL EXPANSION & CONTRACTION, DEFLECTION FOR EXTERNAL FORCES & TORSIONAL LOADS,SETTLEMENTS,TEMPORORY LOADS DURING CONSTRUCTION AND ALL OTHER RELEVANT FORCES. SUPPORTING CALCULATIONS IN RESPECT OF PREDICTION OF AXIAL SHORTENING OF COLUMNS & WALLS TOGETHER WITH PROPOSALS FOR COMPENSATION TO BE SUBMITTED FOR ENGINEERS APPROVAL.

A13. THERE SHALL BE NO REVISIONS TO THE TOLERANCES TO CATER FOR IN INADEQUATE CONTROL BY THE CONTRACTOR.

A15. THE CONTRACTOR SHALL PRODUCE SHOP/WORKING DRAWINGS AND SCHEDULES AS NECESSARY FOR COMPLETION OF THE WORKS AND RECORD DRAWINGS OF THE AS BUILT WORKS, ALL FOR THE ENGINEER'S APPROVAL.

A15. ABBREVIATIONS

APPROX -	APPROXIMATELY
CJ -	CONSTRUCTION JOINT
CONC -	CONCRETE
DIA -	DIAMETER
D -	DEPTH
DWG -	DRAWING
EGL -	EXISTING GROUND LEVEL
EJ -	EXPANSION JOINT
EXTG -	EXISTING
FFL -	FINISHED FLOOR LEVEL
FGL -	FINISHED GROUND LEVEL
GA -	GENERAL ARRANGEMENT
GL -	GROUND LEVEL
GWT -	GROUND WATER TABLE
IJ -	ISOLATION JOINT
m -	METRE
LG -	LONG
MAX -	MAXIMUM
MIN -	MINIMUM
MJ -	MOVEMENT JOINT
mm -	MILLIMETRES
No -	NUMBER
NTS -	NOT TO SCALE
P/O -	PERMANENT OPENING
RAD -	RADIUS
RC -	REINFORCED CONCRETE
REF -	REFER
REQD -	REQUIRED
REV -	REVISION
SOP -	SETTING OUT POINT
SQ -	SQUARE
SSL -	STRUCTURAL SLAB LEVEL
STD -	STANDARD
TBC -	TO BE CONFIRMED
THK -	THICKNESS
TJ -	TRANSVERSE JOINT
TW -	THICKNESS OF WALL
TYP -	TYPICAL
T.O.P. -	TOP OF PARAPET
UNO -	UNLESS NOTED OTHERWISE
W -	WIDE

B. CONCRETE GENERAL ARRANGEMENT NOTES

B1. REFER TO CONSULTANT'S SPECIFICATION FOR CONCRETE - CAST IN PLACE CONCRETE FOR CONCRETE GRADES. FREE WATER/CEMENT RATIO IS INCLUSIVE OF ALL CEMENTITIOUS MATERIAL INCLUDING SILICA FUME AND GGBFS. SUBJECT TO SATISFACTORY COMPLETION OF STRENGTH AND DURABILITY TESTS, THE CONSULTANT MAY GIVE APPROVAL TO REDUCING THE CEMENTITIOUS MATERIAL.

CONCRETE GRADES:-

a). Superstructure Slabs & Beams	40	N/mm2
b). Columns & Walls	50	N/mm2
c). Slab on Grade	25	N/mm2
d). Blinding /Screed Concrete	15	N/mm2
e). PT Slabs	40	N/mm2
f). Raft slab, (na) and other buried elements	50	N/mm2
g). Piles	-	N/mm2
h). All other structural elements.	-	N/mm2

ALL CONCRETE FOR WATER TANKS, SETTLING TANKS AND SLUDGE TANKS (INCLUDES WALLS, FLOOR SLAB AND COVER SLAB) SHALL BE WATERPROOF / WATERTIGHT CONCRETE AND SHALL INCLUDE A HYDROPHOBIC PORE-BLOCKING ADMIXTURE TO REDUCE ABSORBENCY OF THE CONCRETE AND PROVIDE CORROSION RESISTANT CONSTRUCTION. THE ADMIXTURE SYSTEM SHALL BE APPLIED STRICTLY IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS AND TO THE APPROVAL OF THE ENGINEER.

ALL INTERNAL CONCRETE SURFACES INSIDE THE WATER TANK SHALL BE LINED WITH 4MM THICK GRP LAYER AS PER SPECIFICATIONS NO TOXIC APPROVED BY ENGR.

DURABILITY REQUIREMENTS:-
ALL GRADES OF CONCRETE EXPOSED OR IN CONTACT WITH SOIL.

TEST	RESULT
CHLORIDE PERMEABILITY TO AASHTO T277 (COULOMBS)	1000
WATER PERMEABILITY TO DIN 1048 (MM)	8
INITIAL SURFACE ABSORPTION TO BS 1881 PART-5 (ml/m /sec.)	0.15
30 MIN. WATER ABSORPTION TO BS 1881 PART 122 (%)	1.5
MAX. DRYING SHRINKAGE TO BS 812 PART 120 (%)	0.04

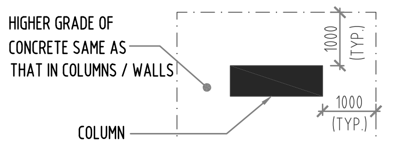
B2. FORMED SURFACES SHALL BE THE FOLLOWING FINISH TYPES AS DEFINED BY THE SPECIFICATION.
EXPOSED COLUMNS & WALLS : TYPE C (SUPERIOR)
ALL OTHER FORMED SURFACES : TYPE B (NORMAL)
CONCRETE IN CONTACT WITH SOIL : TYPE A (ROUGH)

SLIPFORMED CONCRETE FINISHES
WHERE SLIPFORMING IS USED, COMPARABLE FINISHES ARE
CLASS F2 : NORMAL FINISHES
CLASS F3 : SUPERIOR FINISH

B3. UNFORMED SURFACES SHALL RECEIVE THE FOLLOWING FINISH TYPES AS DEFINED IN THE SPECIFICATION.
BLINDING :TF (TAMPED)
SLABS (TO RECEIVE SCREED) :SF (STEEL FLOAT)
SLABS WITHOUT SCREED :PF (POWERFLOAT)
ALL AREAS TO RECEIVE A POWERFLOATED OR BRUSHED FINISH SHALL HAVE AN APPROVED SURFACE HARDENER AND DUST INHIBITOR APPLIED UNLESS APPROVED OTHERWISE.

B4. FOR CLARITY, CHAMFERS ARE NOT SHOWN. REFER STANDARD DETAILS

B5. IN CASE THE GRADE OF CONCRETE IN COLUMNS / WALLS IS HIGHER THAN THAT OF THE SURROUNDING SLAB / BEAMS AT ANY FLOOR LEVEL BY 15N/m2, THE HIGHER GRADE OF CONCRETE SHALL BE POURED IN THE SLAB / BEAMS AROUND THE COLUMNS / WALLS LOCATIONS AS SHOWN BELOW.



CONCRETE SURFACE TREATMENT:

B5. UNLESS NOTED OTHERWISE ALL CONCRETE ELEMENTS BELOW GROUND SHALL BE APPLIED WITH TWO COATS OF BITUMINOUS PAINT.

B6. ALL CONCRETE FACES CONCEALED BENEATH THE CLADDING SHALL BE TREATED WITH A SINGLE COAT OF WATERBASED BITUMEN EMULSION OF APPROVED QUALITY.

B7. UNLESS NOTED OTHERWISE ALL CONCRETE SURFACES (COLUMNS, WALLS, SLABS,BEAMS, SOFFITS ETC) SHALL BE TREATED WITH A PENETRATING SILANE SILOXANE PRIMER OVERLAID BY AN ACRYLIC PROTECTIVE COATING OF APPROVED QUALITY. APPLICABLE TO ALL CONCRETE SURFACES THAT DO NOT HAVE A FINISH IDENTIFIED IN THE SCHEDULE OF INTERNAL FINISHES OR HAVING A WATER PROOFING APPLICATION IDENTIFIED ON THE DRAWINGS / SPECIFICATION.

B8. THE SIZES AND POSITIONS OF HOLES AND OPENINGS FOR SERVICES ARE INDICATIVE ONLY AND SHALL BE CONFIRMED BY THE CONTRACTOR FOR THE INCLUSION IN THE DETAILED SHOP DRAWINGS. ALL HOLES AND OPENINGS SHALL BE POSITIONED TO AVOID MAJOR STRUCTURAL ELEMENTS.

B9. FOR ADDITIONAL REBATES, REFER TO THE ARCHITECT'S DRAWINGS.

B10. ABBREVIATIONS

P -	PERMANENT OPENING
T -	TEMPORARY OPENING
TOC -	TOP OF CONCRETE
B -	BEAM
C -	COLUMN
W -	WALL
F -	PAD FOOTING
L -	LINTEL
RW -	RETAINING WALL
S -	SLAB
SF -	STRIP FOOTING

C. CONCRETE REINFORCEMENT NOTES

C1. UNLESS NOTED OTHERWISE, MINIMUM COVER TO ALL REINFORCEMENT (INCLUDING LINKS) SHALL BE AS FOLLOWS: 75mm
CONCRETE CAST AGAINST SOIL
CONCRETE IN RETAINING WALLS, FOUNDATION & BURIED SLABS. (PROTECTED WITH WATER PROOFING) 50mm
40mm
EXTERNAL FACES 40mm
INTERNAL FACES
EXTERNAL CONCRETE ABOVE GROUND 40mm
INTERNAL CONCRETE: 35mm
FACES ON BUILDING PERIMETER 35mm
SLABS 30mm
BEAMS
COLUMNS & WALLS

C2. REINFORCEMENT ABBREVIATIONS

T	- TOP
B	- BOTTOM
H	- HORIZONTAL
V	- VERTICAL
NF	- NEAR FACE
FF	- FAR FACE
EF	- EACH FACE
BW	- BOTH WAYS
PRS	- PAIRS
ABR	- ALT. BARS REVERSED
ALT	- ALTERNATE
STG	- STAGGERED
TYP.	- TYPICAL
UBAR	- "U" SHAPED REINFORCING BAR

C3. THE DESIGNATION T DENOTES DEFORMED HIGH YIELD STEEL BARS GRADE 500.(YIELD STRENGTH NOT LESS THAN 460 Mpa) COMPLYING TO THE REQUIREMENTS OF ASTM A615.

C4. WHERE REINFORCEMENT IS ASSIGNED LAYER NUMBERS, LAYER 1 IS THAT NEAREST TO THE FACE OF THE CONCRETE.

C5. INSERTS SHALL BE PROVIDED AS NECESSARY, INCLUDING SUPPORT FOR MASONRY AND SERVICES. REINFORCEMENT SHALL BE ADJUSTED TO SUIT. PLASTIC TAPE SHALL BE USED TO PREVENT CONTACT OF DISSIMILAR METALS. ADDITIONAL REINFORCEMENT FOR LOCAL OVERSTRESSING SHALL TO BE PROVIDED AROUND / THROUGH THE INSERT PLATE ANCHORS AS DIRECTED BY THE ENGINEER ON SITE.

C6. REINFORCEMENT DETAILS SHOWN ARE INDICATIVE FOR THE PREPARATION OF THE CONTRACTOR'S WORKING DRAWINGS. THE CONTRACTOR SHALL PROVIDE DETAILED SHOP DRAWINGS AND SCHEDULES OF THE REINFORCEMENT FOR THE ENGINEER'S APPROVAL IN ACCORDANCE WITH THE CONCRETE SPECIFICATION. RC DETAILING SHALL BE TO THE STANDARD METHOD OF DETAILING STRUCTURAL CONCRETE, PUBLISHED BY THE INSTITUTION OF STRUCTURAL ENGINEERS & THE CONCRETE SOCIETY, AUGUST 1989 OR LATEST APPLICABLE AS APPROPRIATE. SHAPE CODES SHALL BE TO BS 8666 (2005), SPECIFICATION FOR BENDING DIMENSIONS AND SCHEDULING OF REINFORCEMENT FOR CONCRETE.

C7. THE CONTRACTOR SHALL SUBMIT PROPOSALS FOR SPACERS, CHAIRS, ETC. FOR THE CONSULTANT'S APPROVAL.

C8. LAPS SHALL BE A MINIMUM OF 50 TIMES OF THE SMALLEST DIAMETER UNLESS NOTED OTHERWISE. CORNER BARS OF COLUMNS AND BEAMS WHERE THE COVER IS LESS THAN 2xBAR DIA AND THE SPACING OF BARS LESS THAN 75 mm LAPS SHALL BE 66 TIMES THE BAR DIAMETER. WELDING OF REBARS NOT PERMITTED.

C9. REINFORCEMENT DETAILING SHALL COMPLY WITH THE REQUIREMENTS OF SECTION 1921.8 OF THE UNIFORM BUILDING CODE (UBC-1997).

C10. REBAR COUPLERS TO BE USED FOR REBARS T32 AND HIGHER DIAMETER, STAGGERED (50% OF NUMBERS, 50 DIA APART.) TESTED AND APPROVED AS PER SPEC. AND BY ENGINEER / AUTHORITY.

C10. TABLE(1)—MINIMUM DIAMETER OF BEND

BAR SIZE	MINIMUM DIAMETER OF BEND
T10 THROUGH T25	6 db
T28 THROUGH T32	8 db

STANDARD HOOKS FOR REINFORCEMENT OTHER THAN STIRRUPS, TIES, AND HOOPS.

6.2 DEVELOPMENT LENGTH, Ld
UNLESS NOTED OTHERWISE ON THE DRAWINGS, DEVELOPMENT LENGTH (Ld) FOR Fcu=40 MPA AND FY=460 MPA SHALL BE AS FOLLOWS :

TABLE (2)— DEVELOPMENT LENGTH OF DEFORMED BARS AND DEFORMED WIRE IN TENSION, Ld

BAR DIAMETER, db (mm).	8	10	12	16	20	25	32
CLEAR SPACING OF BARS BEING DEVELOPED OR SPLICED NOT LESS THAN db, CLEAR COVER NOT LESS THAN db, STIRRUPS OR TIES THROUGHOUT Ld NOT LESS THAN THE CODE MINIMUM OR CLEAR SPACING OF BARS BEING DEVELOPED OR SPLICED NOT LESS THAN 2db AND CLEAR COVER NOT LESS THAN db	360	450	540	720	910	1140	1460
OTHER CASES	440	550	660	880	1370	1710	2180

TABLE (3)— DEVELOPMENT LENGTH OF DEFORMED BARS IN COMPRESSION, Ld

BAR DIAMETER, db (MM)	8	10	12	16	20	25	32
DEVELOPMENT LENGTH, Ld (MM)	200	250	300	400	500	625	800

D. BLOCK WORK

(ALL HOLLOW AND SOLID BLOCK SHALL COMPLY WITH THE REQUIREMENT OF LOCAL ORDER NO.44 OF LOCAL AUTHORITY AND ALL AMENDMENTS AS APPLICABLE) TYPE & LOAD OF THE PARTITIONS TO BE CHECKED WITH THE ARCH. DRAWING

SL.No:	TYPE OF BLOCK	DENSITY (MAX.)	USAGE	AREA
1.	SOLID BLOCK	1800 kg/m^3	BELOW GROUND FLOOR	INTERNAL/EXTERNAL
2.	AAC BLOCK	550 kg/m^3	ALL FLOORS ABOVE GROUND FLOOR	INTERNAL/EXTERNAL
3.	HOLLOW BLOCK	1100 kg/m^3	WET AREAS 2.1m HIGH (MAX.)	INTERNAL

D1. ALL MASONRY WORKS AROUND THE LIFT ARE REINFORCED CONCRETE BLOCK. (FOR DETAILS REFER TO TYPICAL DETAIL IN S103

D2. BLOCK FOR LOAD BEARING WALLS SHALL BE IN SOLID BLOCKS

D3. BLOCK WORKS ABOVE ROOF AND IN BASEMENT FLOORS SHALL BE OF HOLLOW CONCRETE BLOCKS FOR NON LOAD BEARING WALLS

D4. JOINTS BETWEEN NON LOAD BEARING BLOCK WORK AND SUSPENDED SLABS OR BEAMS TO HAVE 12MM COMPRESSIBLE JOINT FILTER

D5. QUALITY OF CONCRETE BLOCKS SHALL BE AS FOLLOWS: MINIMUM CRUSHING STRENGTH AT 28 DAYS WORKING TEST

D6. 1000 GAUGE POLETHENE VAPOUR BARRIER SHALL BE PROVIDE BELOW BASE CONCRETE FOR SLABS AND FOOTING LAID ON DRY COMPACTED GROUND

D7 ONE LAYER OF DAMPER PROOF COURSE (DPC)SHALL BE PROVIDED ALL WALLS TO BS 6398 CLASS D OR E.

D8. WALL PANELS 200mm THICK WALLS SHALL NOT EXEED 14m2 IN AREA LAYER WALL PANELS SHALL BE DIVIDED BY 200x200 mm TIE BEAMS AN / OR STIFFINER COLUMNS AT APPROVED HIGHTS AND SPACING TO COMPLY WITH THIS CONDITION

D9. FOR 300mm TICK WALL ,NO WALL PANEL SHALL BE EXEEDED 30 m2 IN AREA. IN CAESE OF LARGER WALL AREAS,THE WALL SHALL BE DIVIDED BY 300x300mm BEAMS AND /OR COLUMNS AT SUTABLE INTERVALS TO COPLY WITH THIS CONDITION.

D10 CEMENT MORTAR FOR BLOCK WORK: CEMENT TO SAND MIX
USE 1:3 BELOW PLINTH LEVEL (BY VOLUME)
USE 1:4 ABOVE PLINTH LEVEL (BY VOLUME)

D11. FOR ALL CEMENT PLASTERS USE 1:4 CEMENT TO SAND MIX (BY WEIGHT)

D12. W/C RATIO SHAL NOT EXEED (0.5) BY WEIGHT FOR ALL MIXES IN THIS SECTION

D13. PLASTICIERS SHOULD BE USED TO APPROVAL

D14. ALL BLOCK WALL SHALL BE STRENGTHENED WITH STIFFENER COLUMNS AND BEAMS GENERALLY 3.0 m SPACING WITH NOMINAL STEEL AS PER CODE AND DESIGN. ALL BLOCK WALL DESIGN / CALCULATION TO BE PREPARED AND VERIFIED BY CONTRACTOR AND SUBMITTED TO THE ENGINEER FOR APPROVAL

E. STRUCTURAL STEELWORKS NOTES

E1. ALL STRUCTURAL STEELWORK SHALL BE IN ACCORDANCE WITH BS 5950 AND BSEN 10025 OR BSEN 10210 PART1. GRADE S275 (PREVIOUSLY GRADE 43) SHALL BE PROVIDED EXCEPT WHERE NOTED OTHERWISE.
IF SPECIFIED, THE PARTICULAR QUALITY GRADE FOR EACH ELEMENT SHALL BE AS FOLLOWS:–
S275JR FOR MATERIAL OF THICKNESS < 20mm.
S275JO FOR MATERIAL OF THICKNESS >20mm, <40mm.
S275J2G3 FOR MATERIAL OF THICKNESS >40mm, <90mm.
S275JOH FOR HOLLOW SECTIONS THICKNESS <40mm.
S355 OR HIGHER AS PER DESIGN REQD.

E2. ALL STRUCTURAL STEELWORK SHALL BE CORROSION PROTECTED IN ACCORDANCE WITH THE SPECIFICATION.

E3. MEMBERS WHICH ARE CONCRETE ENCASED ARE DENOTED THUS: 305x165 UB 31 (CASED) [EXAMPLE] THESE ELEMENTS REMAIN UNPAINTED AND SHALL BE FREE FROM OIL, GREASE, DIRT, SAND, LOOSE RUST AND MILL SCALE.

E4. CONCRETE CASING SHALL INCORPORATE A SINGLE LAYER OF D49 MESH WITH A MINIMUM LAP OF 300mm.

E5. ALL STEELWORK CONNECTIONS, INCLUDING TYPICAL CONNECTIONS DETAILED ON THE DRAWINGS, SHALL BE DESIGNED BY THE CONTRACTOR IN ACCORDANCE WITH BS 5950. BOLTED CONNECTIONS SHALL, UNLESS NOTED OTHERWISE, UTILISE GRADE 8.8 BOLTS TO BS 3692 OR BS 4190, MINIMUM DIAMETER 16mm, WITH A MINIMUM OF TWO BOLTS PER CONNECTION. BOLTS OF DIAMETER 30mm OR GREATER SHALL BE HIGH STRENGTH FRICTION GRIP IN ACCORDANCE WITH BS 4395 PART 1. CONNECTIONS SHALL BE DESIGNED FOR A MINIMUM FACTORED REACTION OF 75kN IN ANY DIRECTION UNLESS NOTED OTHERWISE.ALL SPLICE JOINTS MUST BE NONSLIP, DESIGNED FOR THE FULL SECTION STRENGTH AT POINT OF CONTRAFLEXURE.
FACTORED BEAM END REACTIONS ARE INDICATED THUS ON PLAN:

EXAMPLE
Fv 80 SHEAR (FORCES IN kN)
Fc 25 AXIAL (FORCES IN kN)
M 105 BENDING MOMENTS (IN kN–m)

FOR BRACING MEMBERS AND LATTICE GIRDER MEMBERS,NO ECCENTRICITY OF THE MEMBER CENTROIDAL AXES MEETING AT A NODE SHALL BE ALLOWED, UNLESS SHOWN OTHERWISE ON THE DRAWING.

E6. UNLESS NOTED OTHERWISE, ALL FILLET WELDS SHALL BE A MINIMUM 6mm LEG LENGTH AND ALL BUTT WELDS SHALL BE FULL PENETRATION. TESTING OF WELDS SHALL BE IN ACCORDANCE WITH THE SPECIFICATION.WELD SYMBOLS ARE TO BS 499. ALL WELDS SHALL BE FULL PROFILE, UNLESS NOTED OTHERWISE.

E7. ALL TUBULAR MEMBERS SHALL, UNLESS NOTED OTHERWISE, BE SEALED AT EACH END WITH A 6mm CAP PLATE CONNECTED BY FILLET WELD. WHERE HOLLOW MEMBERS ARE TO BE GALVANISED, VENT HOLES SHALL BE PROVIDED IN ACCORDANCE WITH THE SPECIFICATION.

E8. PRECAMBER VALUES SHOWN ON THE DRAWINGS. REFER TO THE FABRICATED SHAPE BEFORE LOADS ARE APPLIED.

E9. ALL STEEL STRUCTURE SHALL BE FIRE PROOFED. FIRE RESISTANCE REQUIREMENTS SHALL BE APPLIED IN ACCORDANCE WITH UAE FIRE REGULATIONS.

F. PRECAST CONCRETE SLAB ELEMENTS

F1. PRECAST CONCRETE FLOORING UNITS SHALL BE DESIGNED BY THE CONTRACTOR TO WITHSTAND THE LOADS, GIVEN IN THE SCHEDULE.

F2. THE CONTRACTOR SHALL PROPOSE A VALUE OF PRE CAMBER FOR THE FLOOR PRE CAST OF UNITS, SUBSTANTIATED BY CALCULATION FOR THE ENGINEER'S APPROVAL.

THE PRE CAMBER OF UNITS SHOULD NOT BE MORE THAN SPECIFIED IN THE DRAWINGS.

F3. MANUFACTURERS MUST BE REGISTERED IN RELEVANT AUTHORITY AND ALL MANUFACTURER' S DETAILS MUST BE SUBMITTED FOR ENGINEER'S APPROVAL PRIOR TO MANUFACTURE

F4. UNLESS NOTED OTHER WISE MESH REINFORCEMENT IN ALL TOPPING TO P.C. SLABS SHALL BE A393 TYP. – T10 L BARS REQUIRED TO LAP TO MESH + ANCHOR INTO EDGE BEAMS/WALLS.

G. PRECAST CLADDING ELEMENTS

G1. MINIMUM REINFORCEMENT CONTENT – 0.25 %

G2. MAXIMUM ALLOWABLE CRACK WIDTH – 0.1 MM.

G3. MINIMUM NUMBER OF CONNECTIONS WITH THE MAIN STRUCTURE PER PANEL [b/w FLOOR TO FLOOR.] – 4 NOS. PER EACH 6 M^2 AREA OF THE PANEL.

G4. MINIMUM SIZE OF STAINLESS STEEL PINS – 20 MM. FOR [DOWELS & ANCHORS]

G5. CONTRACTOR TO SUBMIT CALCULATIONS [THIRD PARTY– CERTIFIED] COMPLETE TO THE APPROVAL OF THE ENGINEER.

H. SUBSOIL CONDITION

H1. SOIL INVESTIGATION REPORT No: SR/2103106 REV-03, DATED ON 1st JUNE 2021 AND PREPARED BY AL HAI & AL MUKADDAM FOR GEO TECHNICAL WORKS (L.L.C.)

H2. FOR WATER TABLE LEVELS REFER SOIL INVESTIGATION REPORT.

H3. DEWATERING CAN BE STOPPED AFTER CONSTRUCTION UP TO APPROPRIATE LEVEL TO PROVE THE GRAVITY LOAD IS 1.5 TIMES (MIN.) THAN WATER PRESSURE OR AS PER CODE, AUTHORITY VERIFIED BY CONTRACTOR AND APPROVED BY ENGINEER.

J. TEMPORARY WORKS

THE CONTRACTOR SHALL BE RESPONSIBLE AND LIABLE FOR THE DESIGN OF ALL TEMPORARY WORKS RELATED TO THE CONSTRUCTION PROCESS. THE CONTRACTOR SHALL SUBMIT THE EVIDENCE TO DEMONSTRATE SUITABILITY OF THE TEMPORARY WORKS PRIOR TO THEIR INSTALLATION. CONTRACTOR SHALL PROVIDE A TEMPORARY WORKS COORDINATOR ON SITE WITH THE SPECIFIC RESPONSIBILITY FOR THE ADEQUACY OF ALL TEMPORARY WORKS PROPOSAL.

K. WATERTIGHT CONSTRUCTION:

THE FOLLOWING STRUCTURAL ELEMENTS ARE TO BE CONSTRUCTED IN ACCORDANCE WITH SPECIFICATION.

– TANK WALLS, TANK SLABS
– RAFTS & PILECAPS
– RETAINING WALLS
– LAND SCAPE AREA, SWIMMING POOL/ JACUZZI SLAB AND WALLS

L. EARTHWORKS / DEWATERING:

L1. WHEN EXCAVATING TO FOUNDATION LEVEL CARE SHOULD BE TAKEN NOT TO DISTURB THE UNDERLYING MATERIAL. ALL FOUNDATION EXCAVATIONS SHALL BE INSPECTED AND ANY SOFT SPOTS REMOVED AND REPLACED WITH GRANULAR FILL, WHICH SHALL BE WELL COMPACTED BEFORE CONSTRUCTION PROCEEDS. EXCAVATION SHALL AT ALL TIMES BE CARRIED OUT IN DRY CONDITIONS.

L2. THE CONTRACTOR SHALL CARRY OUT NECESSARY EXCAVATION TO ACHIVE THE FOUNDING LEVELS SHOWN IN THE DRAWINGS OF RAFT SLABS, PILE CAPS, FOOTINGS AND LIFT–PIT SLABS.

L3. THE CONTRACTOR SHALL TAKE FULL ACCOUNT OF THE SOIL INVESTIGATION INFORMATION AND HIS OWN EXPERIENCE TO DESIGN THE NECESSARY TEMPORARY WORKS. THE DESIGN TOGETHER WITH THE SHOP DRAWINGS SHALL BE SUBMITTED FOR THE ENGINEER'S REVIEW PRIOR TO DEWATERING. ALL BACKFILLING SHALL BE COMPACTED TO AT LEAST 95% OF THE MAXIMUM DENSITY AND IN ACCORDANCE WITH THE SPECIFICATIONS.

L4. IT SHALL REMAIN THE CONTRACTOR'S OBLIGATION TO PROVIDE DRY WORKING CONDITIONS BY ANY MEANS HE MIGHT SEE EFFECTIVE. EXCAVATION SHALL BE AT ALL TIMES CARRIED OUT IN DRY CONDITIONS. GROUND WATER SHALL BE MAINTAINED AT LEAST 500mm BELOW THE DEEPEST POINT IN THE EXCAVATION WHILE IT REMAINS OPEN. THE DEWATERING SYSTEM USED SHALL NOT AFFECT THE EXISTING ADJACENT BUILDINGS IN ANY WAY.

M. MANHOLES & SUMPS

GRP LINING SHALL BE APPLIED TO THE FOLLOWING :–
– MANHOLE BOTTOM AND SIDES UP TO 300mm HEIGHT
– FOULWATER / STORMWATER SUMPS
FOR MATERIAL REQUIREMENTS REFER TO ARCHITECTURAL SPECIFICATIONS.

N. CURTAIN WALL DESIGN

N1. THE CONTRACTOR SHALL BE FULLY RESPONSIBLE AND LIABLE FOR THE DESIGN, INSTALLATION AND PERFORMANCE OF THE CURTAIN WALL SYSTEM. THE CONTRACTOR'S DESIGN CALCULATIONS SHALL BE CHECKED AND CERTIFIED BY AN INDEPENDENT THIRD PARTY BEFORE SUBMISSION TO THE ENGINEER TO DEMONSTRATE THE SUITABILITY OF THE SYSTEM FOR ITS INTENDED USE.

N2. THE CURTAIN WALL SHALL INCORPORATE SUFFICIENT FLEXIBILITY TO ACCOMMODATE ALL ANTICIPATED MOVEMENTS IN THE STRUCTURE, INCLUDING THOSE DUE TO THERMAL EFFECTS, LATERAL MOVEMENTS DUE TO WIND OR SEISMIC LOAD AND AXIAL SHORTENING OF COLUMNS. MAX. INELASTIC STOREY DRIFT FOR THE SEISMIC CASE IS 25mm IN ANY DIRECTION BETWEEN TYPICAL FLOORS. THE CURTAIN WALL DESIGN SHALL CONFORM IN ALL RESPECTS TO THE SEISMIC DESIGN REQUIREMENTS OF UBC 1997.

N3. WHERE SHOWN ON THE DRAWINGS, THE CONTRACTOR IS REQUIRED TO DESIGN & ERECT STEEL MEMBERS SUPPORTING THE CURTAIN WALL AS PER LOADS SPECIFIED IN NOTE – A7 CODE REFERENCE BY AUTHORITY AND ENGINEER'S APPROVAL.

P.DRAWING NUMBERING SYSTEM

REFER TO DRAWING LIST FOR LIST OF DRAWING – INDEX AND DETAIL.

Q.POST–TENSIONED CONCRETE WORKS.

Q1. THE SIZE & THICKNESS OF BEAMS & SLABS SHOWN ON THE FLOOR PLANS ARE INDICATIVE

Q2. POST-TENSIONED SLABS ARE TO BE DESIGNED AND DETAILED BY THE CONTRACTOR, WITH CALCULATIONS AND WORKING DRAWINGS SUBMITTED TO THE ENGINEER FOR REVIEW AND NECESSARY AUTHORITY APPROVAL.

Q3. A SPECIALIST SUBCONTRACTOR LICENSED IN DUBAI AND APPROVED BY LOCAL AUTHORITY SHALL UNDER TAKE THE DESIGN AND IMPLEMENTATION OF PT SLABS & BEAMS. AND SHOULD SUBMIT THE DESIGN WARANTY

Q4. POST-TENSIONED WORKS INCLUDING DESIGN, DETAILING, INSTALLATION, MATERIALS, GROUTING AND WORKMANSHIP SHALL CONFORM TO THE REQUIREMENTS OF BS8110, CONCRETE SOCIETY TECHNICAL REPORT No. 43, DRAWINGS AND SPECIFICATION.

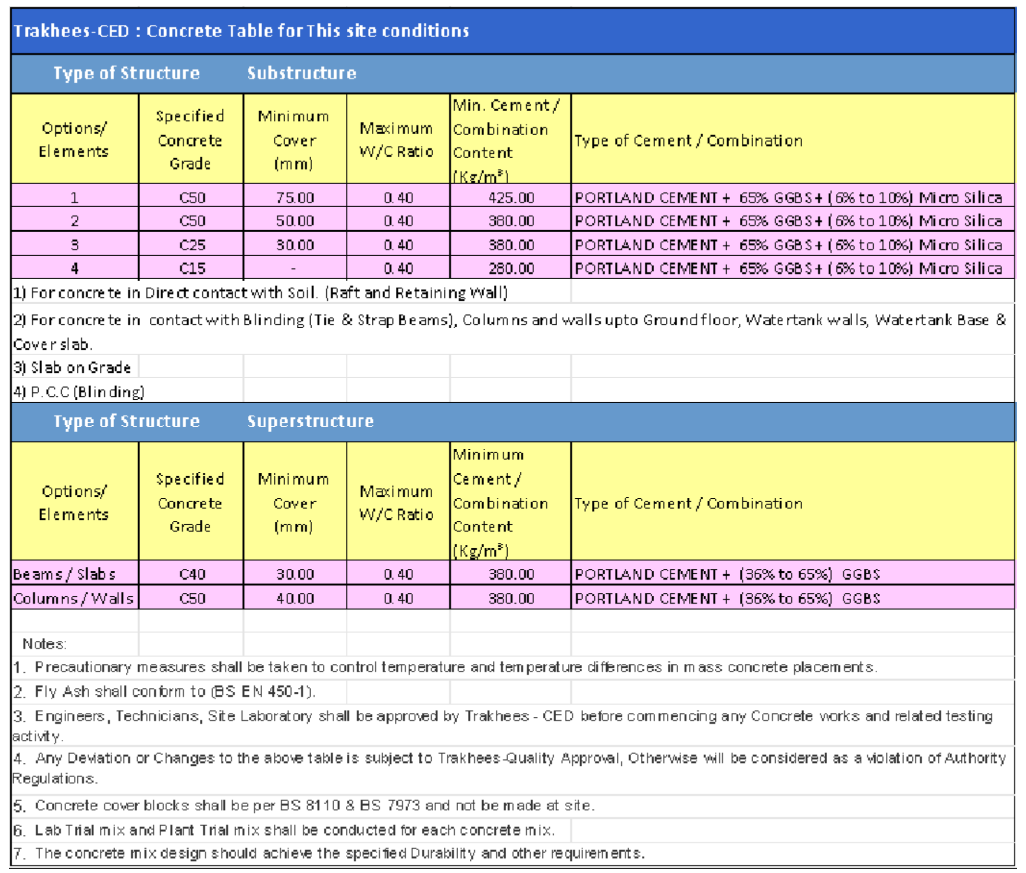
Q5. THE DESIGN SHALL BE IN ACCORDANCE WITH THE PARAMETERS GIVEN IN THE GENERAL NOTES DRAWING FOR THE SUPERIMPOSED DEAD LOADS & LIVE LOADS OF GIVEN, SHALL BE SUBMITTED TO THE ENGINEER FOR APPROVAL.

DEFLECTION LIMIT AS FOLLOWS
LONG TERM L/250 UP TO A MAXIMUM OF 20mm (L/125 CANTILEVER)
INCREMENTAL L/500 UP TO A MAXIMUM OF 15mm (L/250 CANTILEVER)
FULL SET OF CALCULATIONS, DRAWINGS AND DETAILS ARE TO BE PROVIDED TO THE ENGINEER APPROVAL.

Q6. SCOPE OF WORK FOR THE POST-TENSIONING SUB-CONTRACTOR SHALL INCLUDE:

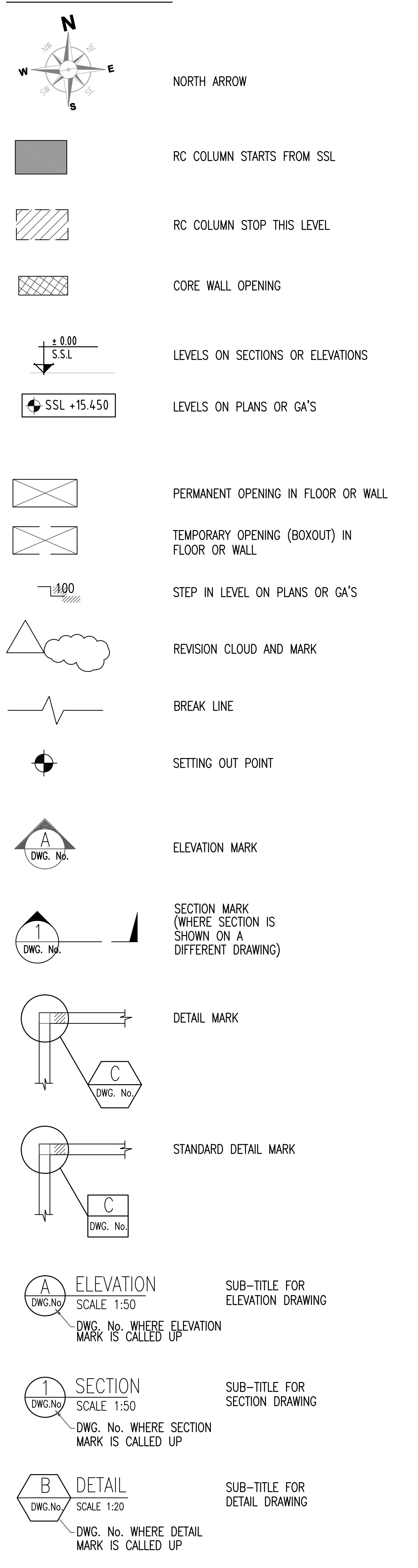
- DESIGN AND DETAILING TO CONSTRUCTION STANDARDS OF ALL POST-TENSIONED ELEMENTS.
- PROVIDE SUPERVISION TO SITE ACTIVITIES RELATED TO POST-TENSIONING WORKS.
- DELIVERY OF ALL MATERIALS SUCH AS DUCTS, TENDONS, ANCHORAGES AND EQUIPMENT RELATED TO POST-TENSIONING TO THE SITE.
- PLACEMENT AND FIXING OF ALL DUCTING INCLUDING SUPPORTING CHAIRS. INSTALLATION OF THE TENDONS.
- STRESSING OF THE TENDONS.
- GROUTING OF THE DUCTS.
- ALL NECESSARY TESTING.
- ALL REMEDIAL WORKS OF ANY DEFECTS RELATED TO THE POST-TENSIONING SYSTEM.

Q7. PT STANDARD DETAILS AND NOTE SHOWN IN THE DRAWING (DWG. No: S-009) ARE INDICATIVE .



Trakhees-CED : Concrete Table for This site conditions					
Type of Structure	Substructure				
Options/ Elements	Specified Concrete Grade	Minimum Cover (mm)	Maximum W/C Ratio	Min. Cement / Combination Content (Kg/m^3)	Type of Cement / Combination
1	C50	75.00	0.40	425.00	PORTLAND CEMENT + 65% GGBS + (6% to 10%) Micro Silica
2	C50	50.00	0.40	380.00	PORTLAND CEMENT + 65% GGBS + (6% to 10%) Micro Silica
3	C25	30.00	0.40	380.00	PORTLAND CEMENT + 65% GGBS + (6% to 10%) Micro Silica
4	C15	-	0.40	280.00	PORTLAND CEMENT + 65% GGBS + (6% to 10%) Micro Silica
1) For concrete in Direct contact with Soil. (Raft and Retaining Wall)					
2) For concrete in contact with Blinding (Tie & Strap Beams), Columns and walls upto Ground floor, Watertank walls, Watertank Base & Cover slab.					
3) Slab on Grade					
4) P.C.C (Blinding)					
Type of Structure	Superstructure				
Options/ Elements	Specified Concrete Grade	Minimum Cover (mm)	Maximum W/C Ratio	Minimum Cement / Combination Content (Kg/m^3)	Type of Cement / Combination
Beams / Slabs	C40	30.00	0.40	380.00	PORTLAND CEMENT + (36% to 65%) GGBS
Columns / Walls	C50	40.00	0.40	380.00	PORTLAND CEMENT + (36% to 65%) GGBS
Notes:					
1. Precautionary measures shall be taken to control temperature and temperature differences in mass concrete placements.					
2. Fly Ash shall conform to (BS EN 450-1).					
3. Engineers, Technicians, Site Laboratory shall be approved by Trakhees - CED before commencing any Concrete works and related testing activity.					
4. Any Deviation or Changes to the above table is subject to Trakhees-Quality Approval, Otherwise will be considered as a violation of Authority Regulations.					
5. Concrete cover blocks shall be per BS 8110 & BS 7973 and not be made at site.					
6. Lab Trial mix and Plant Trial mix shall be conducted for each concrete mix.					
7. The concrete mix design should achieve the specified Durability and other requirements.					

STANDARD SYMBOLS


N
W
E
S
NORTH ARROW
RC COLUMN STARTS FROM SSL
RC COLUMN STOP THIS LEVEL
CORE WALL OPENING
± 0.00
S.S.L
LEVELS ON SECTIONS OR ELEVATIONS
SSL +15.450
LEVELS ON PLANS OR GA'S
PERMANENT OPENING IN FLOOR OR WALL
TEMPORARY OPENING (BOXOUT) IN FLOOR OR WALL
100
STEP IN LEVEL ON PLANS OR GA'S
REVISION CLOUD AND MARK
BREAK LINE
SETTING OUT POINT
A
DWG. No.
ELEVATION MARK
1
DWG. No.
SECTION MARK (WHERE SECTION IS SHOWN ON A DIFFERENT DRAWING)
C
DWG. No.
DETAIL MARK
C
DWG. No.
STANDARD DETAIL MARK
A
DWG.No
ELEVATION
SCALE 1:50
DWG. No. WHERE ELEVATION MARK IS CALLED UP
SUB–TITLE FOR ELEVATION DRAWING
1
DWG.No
SECTION
SCALE 1:50
DWG. No. WHERE SECTION MARK IS CALLED UP
SUB–TITLE FOR SECTION DRAWING
B
DWG.No.
DETAIL
SCALE 1:20
DWG. No. WHERE DETAIL MARK IS CALLED UP
SUB–TITLE FOR DETAIL DRAWING

E10. ABBREVIATIONS

CH PL -	CHEQUERED PLATE
CHS -	CIRCULAR HOLLOW SECTION
CSK -	COUNTER SUNK
FPBW -	FULL PENETRATION BUTT WELD
FW -	FILLET WELD
HSFG BOLTS -	HIGH STRENGTH FRICTION GRIP BOLTS
O/A -	OVER ALL
PL -	PLATE
RHS -	RECTANGULAR HOLLOW SECTION
RSA -	ROLLED STEEL ANGLE
RSC -	ROLLED STEEL CHANNEL
RSJ -	ROLLED STEEL JOIST
SHS -	SQUARE HOLLOW SECTION
TOS -	TOP OF STEEL
UB -	UNIVERSAL BEAM
UC -	UNIVERSAL COLUMN
R/F -	REINFORCEMENT

NOTES :
1. THIS DRAWING MAY NOT BE REPRODUCED BY ANY MEANS WITHOUT PRIOR APPROVAL FROM EMSQUARE
2. DRAWINGS ARE NOT TO BE SCALED,STATED DIMENSIONS ONLY ARE TO BE TAKEN.
3. ANY INCONSISTENCY OR DISCRIPANCY IS TO BE BROUGHT TO THE ATTENTION OF & RESOLVED BY THE ARCHITECT/ENGINEER BEFORE THE WORK IS PUT IN HAND.

DATE	DESCRIPTION	REV.	BY
13.04.2026	FOR AUTHORITY APPROVAL	00	MS

CONSULTANT :
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ENGINEERING CONSULTANTS
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OWNER :
G F S REAL ESTATE L.L.C

JOB TITLE :
PROPOSED RESIDENTIAL BUILDING
(B+G+5+ROOF)

PLOT NO. / PARCEL ID: 5310319
BLOCK / COMMUNITY: SAIH SHUAIB 2
DRAWING TITLE : STR. GENERAL NOTES
ARCHITECT : SHAFI
STRUCTURAL ENGR. : Eng.Vincent
DRAWN : MUFADDAL
SCALE :
DATE : 13.04.2026
CHECKED BY : Eng.Vincent
JOB NO : —
DRAWING NO. : S-00_2
REVISION NO. :

PRINTED BY :USER-4.-STRUCTURE GFS B+G+5+ROOF R00.DWG -APRIL/13/2026 - 2:45PM)